

SHAWBURY, United Kingdom

WMO: 034140

Lat: 52.7947N

Lon: 2.6647W

Elev: 72

StdP: 100.46

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.5	-2.9	-6.3	2.2	-2.2	-4.8	2.5	-1.0	14.4	9.5	13.1	9.3	1.6	330	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	25.6	17.9	23.5	16.9	21.7	16.3	18.8	23.4	17.8	21.8	17.0	20.6	4.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.3	20.4	16.2	11.6	19.5	15.5	11.1	18.7	53.5	23.6	50.5	21.9	47.9	20.7	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.2	9.0	DB	-8.4	28.6	2.9	2.0	-10.5	30.0	-12.2	31.2	-13.8	32.4	-15.9	33.8	(4)
(5)			WB	-8.7	21.0	2.8	1.6	-10.8	22.1	-12.4	23.1	-14.0	24.0	-16.1	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	4.4	4.7	6.2	8.3	11.3	14.2	16.2	15.9	13.5	10.5	7.0	4.7	(6)
(7)		DBStd	5.20	3.38	3.24	2.97	2.93	2.88	2.52	2.48	2.35	2.34	3.01	3.30	3.92	(7)
(8)		HDD10.0	831	174	151	122	66	18	1	0	0	2	30	100	167	(8)
(9)		HDD18.3	3150	432	382	377	300	218	127	77	83	146	243	341	423	(9)
(10)		CDD10.0	743	1	2	4	15	59	126	191	182	107	46	8	3	(10)
(11)		CDD18.3	21	0	0	0	0	0	3	10	7	1	0	0	0	(11)
(12)		CDH23.3	192	0	0	0	0	5	32	98	51	5	1	0	0	(12)
(13)		CDH26.7	26	0	0	0	0	0	3	14	9	0	0	0	0	(13)
(14)	Wind	WSAvg	4.2	4.7	4.8	4.6	4.2	4.1	3.9	3.8	3.8	4.1	4.1	4.4	(14)	
(15)	Precipitation	PrecAvg	658	56	42	46	49	56	54	52	60	61	62	64	(15)	
(16)		PrecMax	867	110	111	96	115	173	124	119	128	175	114	135	130	(16)
(17)		PrecMin	455	10	4	7	7	12	12	8	8	2	11	23	9	(17)
(18)		PrecStd	91	26	24	20	24	31	29	28	28	38	28	27	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	13.6	15.9	20.7	24.0	26.9	28.4	27.9	24.0	19.3	15.4	14.1	(19)
(20)			MCWB	11.1	11.0	10.9	13.7	16.5	18.7	18.7	18.1	17.1	15.8	13.5	12.2	(20)
(21)		2%	DB	11.7	12.0	13.7	17.2	21.2	23.5	26.0	24.4	21.2	17.3	14.2	12.7	(21)
(22)			MCWB	9.9	9.9	10.2	11.9	15.1	16.8	17.9	17.7	16.3	14.6	12.4	11.1	(22)
(23)		5%	DB	10.6	10.9	12.3	15.1	18.8	21.3	23.9	22.2	19.4	16.2	13.1	11.6	(23)
(24)			MCWB	8.9	8.9	9.6	10.9	14.0	15.8	17.1	16.9	15.2	13.8	11.4	10.1	(24)
(25)		10%	DB	9.6	9.8	11.2	13.6	16.8	19.4	21.7	20.5	18.0	15.1	12.0	10.4	(25)
(26)			MCWB	8.0	8.0	8.7	10.1	12.7	14.8	16.3	16.0	14.4	13.0	10.3	9.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.4	12.0	14.2	17.6	19.6	20.4	19.8	18.5	16.4	13.9	12.2	(27)
(28)			MCDWB	12.7	12.9	14.1	18.7	22.3	24.6	26.0	24.2	21.9	18.6	15.0	13.4	(28)
(29)		2%	WB	10.2	10.2	10.9	12.6	15.7	17.6	18.8	18.6	17.1	15.2	12.7	11.4	(29)
(30)			MCDWB	11.5	11.8	13.0	16.4	20.2	21.8	24.0	22.5	20.1	16.9	14.0	12.6	(30)
(31)		5%	WB	9.2	9.2	10.0	11.5	14.5	16.4	17.8	17.6	16.0	14.3	11.6	10.3	(31)
(32)			MCDWB	10.4	10.7	11.9	14.6	18.1	20.2	22.3	21.1	18.2	15.8	12.9	11.5	(32)
(33)		10%	WB	8.1	8.2	9.0	10.4	13.3	15.4	16.9	16.8	15.1	13.3	10.6	9.1	(33)
(34)			MCDWB	9.4	9.7	10.9	13.1	16.1	18.8	20.9	19.9	17.4	14.8	11.8	10.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.6	6.2	7.4	8.6	8.9	8.8	9.1	8.7	8.3	7.1	6.1	5.6	(35)
(36)			MCDBR	6.3	6.7	8.9	11.1	12.2	12.6	13.1	11.2	10.5	7.4	6.0	5.9	(36)
(37)			MCWBR	5.8	5.4	6.2	6.7	7.2	6.9	6.4	5.6	6.0	5.2	5.1	5.4	(37)
(38)		5% WB	MCDWR	6.1	6.6	7.8	10.2	10.9	10.8	11.1	9.5	8.7	6.9	6.1	6.0	(38)
(39)			MCWBR	5.8	5.7	5.9	6.6	6.9	6.5	6.2	5.5	5.8	5.5	5.5	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.319	0.335	0.366	0.391	0.396	0.392	0.391	0.376	0.363	0.346	0.321	0.310	(40)	
(41)		taud	2.347	2.386	2.299	2.241	2.267	2.306	2.312	2.371	2.410	2.431	2.402	2.319	(41)	
(42)		Ebn at Noon	661	772	815	839	854	860	853	847	813	748	657	595	(42)	
(43)		Edh at Noon	60	78	103	123	126	122	120	107	92	73	57	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.70	1.35	2.48	3.83	4.78	5.08	4.75	3.89	2.90	1.62	0.83	0.53	(44)	
(45)		RadStd	0.06	0.15	0.27	0.41	0.30	0.53	0.51	0.28	0.23	0.15	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page